Supplementary text

PubMed search terms:

1) “Body mass index” OR body mass index [MeSH] OR BMI OR overweight OR overweight [MeSH] OR obesity OR obesity[MeSH] OR anthropometry OR anthropometry[MeSH] OR fatness OR “body fatness” OR “abdominal fatness” OR “abdominal obesity” OR abdominal obesity[MeSH] OR adiposity OR adiposity[MeSH]

2) Diabetes OR diabetes [MeSH] OR glucose OR glucose [MeSH] OR "medical history" OR medical history[MeSH]

3) “blood pressure” OR blood pressure [MeSH] OR hypertension OR hypertension[MeSH])

4) smoking OR smoking[MeSH] OR cigarette OR cigarette[MeSH] OR cigarettes OR cigarettes[MeSH] smoke OR smoke[MeSH] OR tobacco OR tobacco[MeSH] OR snus OR snuff OR snuff[MeSH] OR "environmental tobacco smoke” OR environmental tobacco smoke[MeSH] OR “passive smoking” OR passive smoking[MeSH] OR “smoking cessation" OR smoking cessation[MeSH]

5) COVID-19 OR SARS-CoV-2) AND (MORTALITY OR DEATH)

6) 1 OR 2 OR 3 OR 4

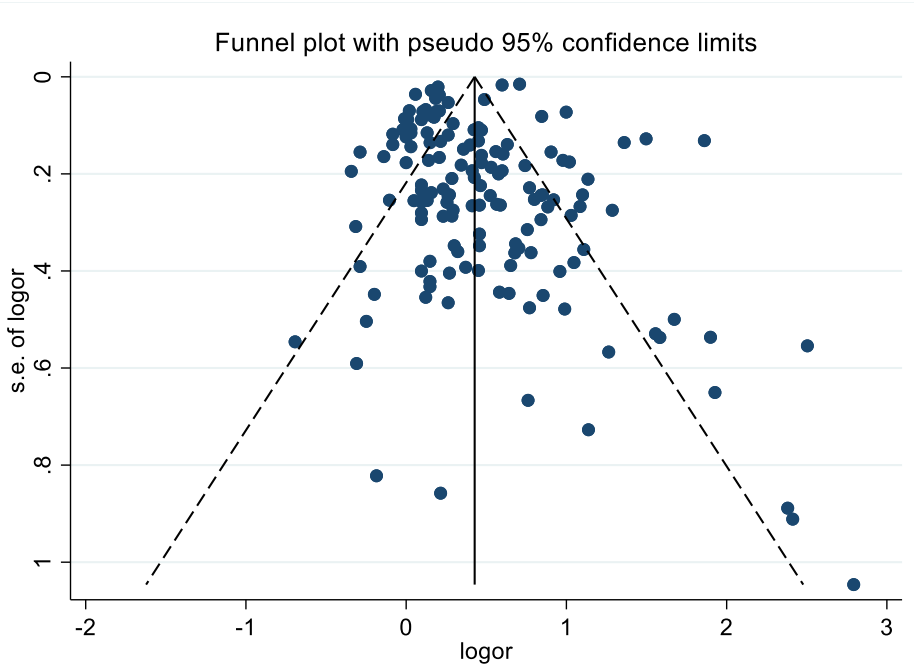
8) 6 AND 5

Embase search terms:

☐	1	(body mass index or bmi or overweight or obesity or anthropometry or fatness or body fatness or abdominal fatness or abdominal obesity or adiposity or diabetes or blood glucose or medical history or blood pressure or hypertension or smoking or cigarette or cigarettes or smoke or tobacco or snus or snuff or environmental tobacco smoke or passive smoking or smoking cessation).ab,ti.	2489416	Advanced
☐	2	body mass index/ or bmi/ or overweight/ or obesity/ or anthropometry/ or fatness/ or body fatness/ or abdominal fatness/ or abdominal obesity/ or adiposity/ or diabetes/ or blood glucose/ or medical history/ or blood pressure/ or hypertension/ or smoking/ or cigarette/ or cigarettes/ or smoke/ or tobacco/ or snus/ or snuff/ or environmental tobacco smoke/ or passive smoking/ or smoking cessation/	2393322	Advanced
☐	3	(COVID-19 or SARS-CoV-2).ab,ti.	66858	Advanced
☐	4	COVID-19/ or SARS-CoV-2/	95	Advanced
☐	5	(mortality or death).ab,ti.	2047722	Advanced
☐	6	mortality/ or death/	1077889	Advanced
☐	7	1 or 2	3256028	Advanced
☐	8	3 or 4	66898	Advanced
☐	9	5 or 6	2342175	Advanced
☐	10	7 and 8 and 9	2095	Advanced

Cochrane library search terms:

#1	Body mass index
#2	anthropometry
#3	obesity
#4	overweight
#5	adiposity
#6	#1 OR #2 OR #3 OR #4 OR #5
#7	Diabetes
#8	glucose
#9	medical history
#10	#7 OR #8 OR #9
#11	blood pressure
#12	hypertension
#13	#11 OR #12
#14	smoking
#15	cigarette
#16	tobacco
#17	#14 OR #15 OR #16
#18	COVID-19
#19	SARS-CoV-2
#20	#18 OR #19
#21	MORTALITY
#22	DEATH
#23	#21 OR #22
#24	(#6 OR #10 OR #13 OR #20) AND #23



Supplementary Figure 1: Funnel plot of diabetes and mortality risk in patients with COVID-19

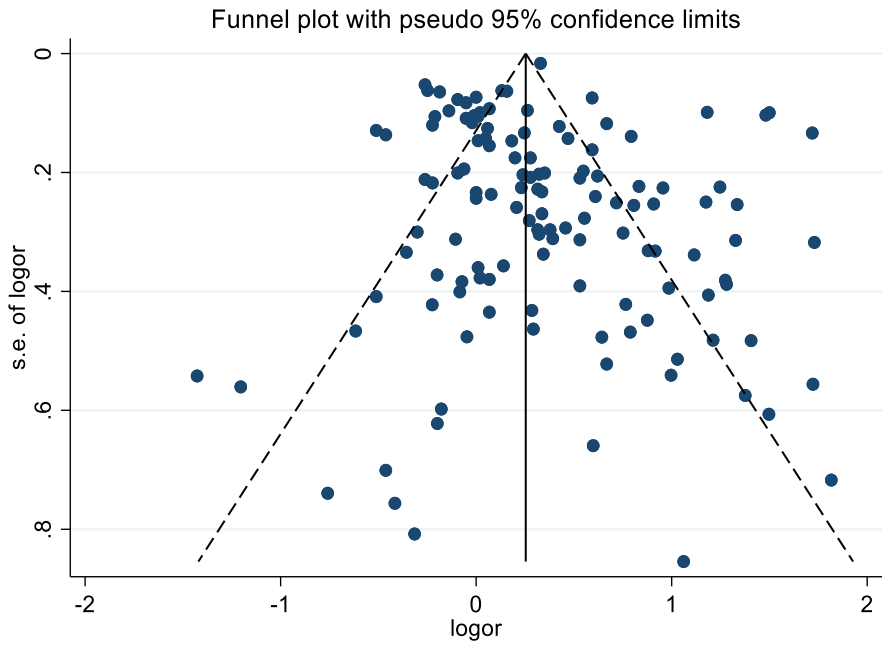
Study omitted	Estimate	[95% Conf. Interval]
Albitar, 2020	1.5293853	1.4358113 1.6290579
Al-Salameh, 2020	1.5442724	1.4494652 1.645281
Hashemi, 2020	1.5388012	1.444257 1.6395345
Atkins, 2020	1.5295212	1.435613 1.6295722
Barbu, 2020	1.5139627	1.4230996 1.6106272
Barron, 2020	1.5295643	1.4380455 1.6269073
Boulle, 2020	1.5274702	1.4342617 1.626736
Bousquet, 2020	1.537884	1.4434227 1.638527
Calmes, 2020	1.5364693	1.4419811 1.6371492
Cao, 2020	1.5384178	1.4440672 1.6389329
Carrasco-Sánchez, 2020	1.5381721	1.4424546 1.6402413
Castelnuovo, 2020	1.5441017	1.4491032 1.645328
Chachkhiani, 2020	1.5411512	1.4464148 1.6420925
Chang, 2020	1.5396796	1.4451315 1.6404138
Chen, 2020	1.5313251	1.4371679 1.631651
Chen, 2020	1.5329227	1.4387097 1.6333052
Chilimuri, 2020	1.5377694	1.4431939 1.6385427
Ciardullo, 2020	1.5410501	1.4461583 1.6421683
Coppelli, 2020	1.5411114	1.446546 1.6418588
Crouse, 2020	1.5296242	1.435763 1.6296216
Cummings, 2020	1.5397375	1.4450227 1.6406603
Dennis, 2020	1.5427595	1.4469318 1.6449338
Desai, 2020	1.5401301	1.4455459 1.6409032
DF de Souza, 2020	1.5309588	1.4368759 1.6312019
Escalera-Antezana, 2020	1.5346913	1.4406917 1.6348242
Esme, 2020	1.5427371	1.4472408 1.6445348
Fang, 2020	1.5347261	1.4405845 1.63502
Fernandez-Cruz. , 2020	1.5323932	1.4382639 1.632683
Fried, 2020	1.5448198	1.4499104 1.6459421
Galloway, 2020	1.5412358	1.4462699 1.6424375
Ghany, 2020	1.5436468	1.448668 1.6448528
Grasselli, 2020	1.5424429	1.4473172 1.6438209
Gu, 2020	1.541069	1.4463067 1.6420403
Gu, 2020	1.5362892	1.4419804 1.6367661
Guisado-Vasco, 2020	1.5346212	1.4403979 1.6350082
Gupta, 2020	1.5425971	1.4475741 1.6438575
Gutierrez, 2020	1.5402496	1.441107 1.6462128
Haase, 2020	1.541796	1.4469641 1.6428429
Harrison, 2020	1.5433826	1.4483094 1.6446967
Narain, 2020	1.5444776	1.4496069 1.645557

Hu, 2020	1.5414069	1.4467993	1.6422008
Imam, 2020	1.5233965	1.4304692	1.6223607
Kim, 2020	1.5359648	1.4414166	1.6367149
Kim, 2020	1.5423032	1.4471788	1.6436802
Klang, 2020	1.5443149	1.449327	1.6455281
Iaccarino, 2020	1.5368001	1.4421782	1.6376301
Lampasona, 2020	1.5336277	1.4393409	1.6340909
Lee, 2020	1.5330603	1.4390559	1.6332053
Li, 2020	1.5316471	1.4377936	1.6316268
Li, 2020	1.543546	1.4486066	1.6447074
Li, 2020	1.5410111	1.4462956	1.6419294
Liu, 2020	1.537012	1.4427184	1.6374683
Liu, 2020	1.5344501	1.4404455	1.6345894
Lombardi, 2020	1.5391879	1.444409	1.640186
Lunski, 2020	1.5452052	1.4502395	1.6463894
Maddaloni, 2020	1.5388812	1.4444	1.6395427
Mallow, 2020	1.5415205	1.4459832	1.64337
McCarty, 2020	1.5362325	1.4418561	1.6367861
Mccullough, 2020	1.5416657	1.4468532	1.6426914
Mendy, 2020	1.5371674	1.4427798	1.63773
Menezes-Soares, 2020	1.535192	1.4406074	1.6359866
Mikami, 2020	1.5455133	1.4505748	1.6466656
Mirani, 2020	1.5397638	1.4450005	1.6407416
Moon, 2020	1.530614	1.436552	1.6308349
Moon, 2020	1.5351865	1.4409643	1.6355697
Munblit, 2020	1.5340241	1.4396216	1.634617
Nachega, 2020	1.5413967	1.446624	1.6423783
Nakeshbandi, 2020	1.5433714	1.4483217	1.644659
Anzola, 2020	1.5396569	1.445129	1.640368
Nicholson, 2020	1.5342927	1.4400049	1.6347542
Nogueira, 2020	1.5475264	1.4526211	1.6486322
Palaiodimos, 2020	1.539917	1.4453287	1.6406955
Pan, 2020	1.5393479	1.444692	1.6402055
Perkin, 2020	1.5366874	1.4421818	1.637386
Petrilli, 2020	1.5445431	1.4495956	1.6457098
Pettit, 2020	1.542518	1.4479332	1.6432815
Poblador-Plou, 2020	1.538321	1.4433994	1.6394849
Portoles, 2020	1.535696	1.4411083	1.636492
Rastad, 2020	1.5370438	1.4424915	1.6377939
Recinella, 2020	1.5426378	1.4479485	1.6435194
Reilev, 2020	1.5376048	1.4427335	1.6387147

Rivera-Izquierdo, 2020	1.5335327 1.4392818 1.6339558
Rodrıguez-Molinero, 2020	1.5314742 1.4374141 1.6316895
Rossi, 2020	1.5375675 1.4428757 1.6384736
Rottoli, 2020	1.5418841 1.4470898 1.6428881
Rozaliyani, 2020	1.54023 1.4454752 1.6411963
Salacup, 2020	1.5383996 1.4439025 1.6390811
Salazar, 2020	1.5320424 1.4381227 1.6320957
Sands, 2020	1.5378478 1.4430354 1.6388896
Santos, 2020	1.5433739 1.4477333 1.6453327
Seiglie, 2020	1.5359882 1.441622 1.6365315
Shah, 2020	1.5382174 1.4436157 1.6390185
Shang, 2020	1.5354923 1.4411708 1.6359869
Shi, 2020	1.5378044 1.443289 1.6385093
Smith, 2020	1.5391345 1.4442916 1.6402056
Sousa, 2020	1.5397989 1.445085 1.6407204
Stony Brook COVID-19, 2020	Research Consortium 1.5362756 1.441714 1.6370395
Talavera, 2020	1.5393621 1.444719 1.6402053
Tartof, 2020	1.5421256 1.447147 1.6433378
Turcotte, 2020	1.5361555 1.4419419 1.6365249
Wollenstein-Betech, 2020	1.5441164 1.4472077 1.6475143
Wu, 2020	1.5398521 1.4451839 1.6407217
Wu, 2020	1.5378178 1.4433225 1.6384996
Xie, 2020	1.5365462 1.442167 1.6371019
Yahyavi, 2020	1.5406708 1.4456849 1.6418976
Yan, 2020	1.5409604 1.4461825 1.6419497
Yan, 2020	1.5381222 1.4434472 1.6390067
Yang, 2020	1.5376757 1.4430538 1.6385021
Yazdanpanah, 2020	1.5360827 1.4417021 1.636642
Yu, 2020	1.5335183 1.43924 1.6339724
Zandkarimi, 2020	1.5342196 1.4398625 1.6347601
Zhang, 2020	1.5332291 1.4392681 1.6333241
Zhou, 2020	1.53381 1.4396418 1.634138
Zhu, 2020	1.5385867 1.4437524 1.6396505
Zimering, 2020	1.5406566 1.4461459 1.6413438
An, 2020	1.5363024 1.441656 1.6371624
Almazeed, 2020	1.5392737 1.4448762 1.6398386
Bepouka, 2020	1.5390179 1.4444937 1.6397277
Biscarini, 2020	1.5398027 1.4451042 1.6407068
Abohamr, 2020	1.5307534 1.4367411 1.6309172
Czernichow, 2020	1.5404149 1.4453428 1.6417407
Docherty, 2020	1.5435574 1.4491569 1.6441072

Foy, 2020	1.545491	1.4505337 1.6466644
Harmouch, 2020	1.5434103	1.4485663 1.6444643
Hewitt, 2020	1.544011	1.4490118 1.6452384
Ioannou, 2020	1.5431746	1.4480621 1.6445346
Kabarriti, 2020	1.5422004	1.4469972 1.6436675
Kolhe, 2020	1.5418558	1.4469479 1.6429888
Li, 2020	1.5343466	1.4403477 1.6344799
Meng, 2020	1.5472357	1.452306 1.6483704
Namendys-Silva, 2020	1.5329634	1.4388483 1.6332345
Neumann-Podczaska, 2020	1.5356869	1.4414098 1.6361301
Nie, 2020	1.5352678	1.4409221 1.6357909
Núñez-Gil, 2020	1.5301256	1.4361231 1.6302811
Park, 2020	1.5341833	1.4398587 1.6346871
Parker, 2020	1.5368429	1.4424789 1.6373801
Rosenberg, 2020	1.5379175	1.4430023 1.6390758
Shi, 2020	1.5396239	1.445085 1.6403475
Tehrani, 2020	1.5389986	1.4444335 1.6397549
Wang, 2020	1.5375615	1.4428405 1.6385009
Arshad, 2020	1.544768	1.4498014 1.6459552
Berenguer, 2020	1.5419745	1.4467498 1.6434668
Ciceri, 2020	1.5382968	1.4435889 1.6392183
Deng, 2020	1.5205251	1.428224 1.6187912
Wang, 2020	1.5330503	1.4390513 1.6331892
Combined	1.5379736	1.4437299 1.6383693

Supplementary Figure 2: Influence analysis of diabetes and mortality risk in patients with COVID-19



Supplementary Figure 3: Funnel plot of hypertension and mortality risk in patients with COVID-19

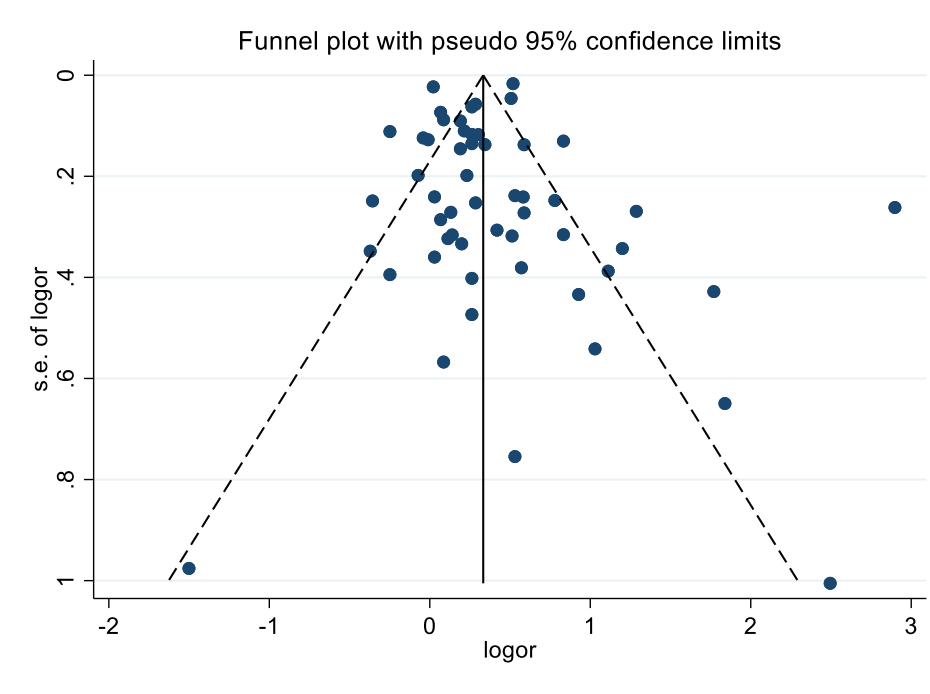
Study omitted	Estimate	[95% Conf. Interval]
Albitar, 2020	1.409848	1.2944931 1.5354825
Almazeedi, 2020	1.420794	1.3044698 1.5474913
An, 2020	1.4202995	1.3034456 1.5476295
Anzola, 2020	1.4250737	1.3083675 1.5521902
Arshad, 2020	1.42682	1.3096998 1.5544138
Atkins, 2020	1.4181341	1.3015437 1.5451684
Barbu, 2020	1.4032818	1.2905699 1.5258374
Bepouka, 2020	1.4205649	1.3040783 1.5474566
Berenguer, 2020	1.4144012	1.2976918 1.5416069
Biscarini, 2020	1.4260321	1.3090302 1.5534917
Boulle, 2020	1.4235396	1.3060689 1.551576
Bousquet, 2020	1.4184818	1.3022381 1.545102
Brouns, 2020	1.4248693	1.3082318 1.5519058
Calmes, 2020	1.4182589	1.3018024 1.5451334
Cao, 2020	1.4202471	1.3040493 1.5467987
Carrasco-Sánchez, 2020	1.4225355	1.3039615 1.5518919
Castelnuovo, 2020	1.4258839	1.3085872 1.5536945
Chachkhiani, 2020	1.4165195	1.3003861 1.5430244
Chen, 2020	1.4063388	1.2914803 1.5314124
Chen, 2020	1.4216918	1.3051524 1.5486373
Chen, 2020	1.4113356	1.2955974 1.5374128
Chilimuri, 2020	1.4177847	1.3014145 1.5445606
Ciardullo, 2020	1.4191939	1.3024282 1.5464277
Ciceri, 2020	1.4102607	1.2947223 1.5361094
Coppelli, 2020	1.4210031	1.3044831 1.5479311
Crouse, 2020	1.4099303	1.2945702 1.5355703
Cummings, 2020	1.416917	1.3006244 1.5436076
Czernichow, 2020	1.4236165	1.3062171 1.5515677
DE VITO, 2020	1.4222373	1.3059056 1.548932
DF de Souza, 2020	1.4171228	1.3003109 1.5444287
Deng, 2020	1.3963305	1.2868365 1.515141
Desai, 2020	1.4228125	1.3060952 1.54996
Du, 2020	1.4107752	1.295385 1.5364441
Escalera-Antezana, 2020	1.4110782	1.2955806 1.5368721
Esme, 2020	1.4221164	1.3035688 1.5514448
Fang, 2020	1.4226052	1.3060334 1.5495815
Foy, 2020	1.422677	1.3055518 1.55031
Fried, 2020	1.4274895	1.3108877 1.554463
Galloway, 2020	1.4198248	1.3028002 1.5473611

Gao, 2020	1.4137607	1.2978215	1.5400571
Ghany, 2020	1.4183894	1.3019762	1.5452114
Giacomelli, 2020	1.4181474	1.3018019	1.5448911
Grasselli, 2020	1.423923	1.3065236	1.5518717
Gu, 2020	1.4185693	1.3020047	1.5455697
Guisado-Vasco, 2020	1.4154416	1.2991362	1.5421594
Gupta, 2020	1.4226934	1.3054665	1.5504467
Gutierrez, 2020	1.4256142	1.2931122	1.5716934
Haase, 2020	1.4236912	1.3067129	1.5511414
Harmouch, 2020	1.4136345	1.2976446	1.5399922
Hashemi, 2020	1.4149497	1.2990468	1.5411937
Hewitt, 2020	1.4244795	1.3071433	1.5523484
Hu, 2020	1.420043	1.3035676	1.5469255
Iaccarino, 2020	1.4224738	1.3056414	1.5497606
Imam, 2020	1.3971469	1.2870134	1.5167049
Ioannou, 2020	1.4247838	1.3071853	1.5529618
Kabarriti, 2020	1.4241887	1.3062947	1.5527228
Kim, 2020	1.4190946	1.3024367	1.5462012
Kim, 2020	1.4222981	1.3052446	1.5498487
Klang, 2020	1.4207023	1.303695	1.5482111
Lee, 2020	1.4160867	1.3001062	1.5424136
Li, 2020	1.4208537	1.304601	1.5474657
Li, 2020	1.4118376	1.2960577	1.5379604
Liu, 2020	1.4229921	1.3063434	1.5500568
Liu, 2020	1.4283701	1.3117267	1.5553858
Lombardi, 2020	1.4196392	1.3029786	1.5467449
Lopez-Mendez, 2020	1.4123951	1.2968653	1.5382168
Lunski, 2020	1.4306874	1.3140447	1.5576842
MCCULLOUGH, 2020	1.4113107	1.2956346	1.5373144
Maddaloni, 2020	1.4137343	1.2980008	1.5397871
Magleby, 2020	1.4250162	1.308177	1.5522907
Mallow, 2020	1.4274716	1.3114311	1.55378
Matsuzawa, 2020	1.4213893	1.3050824	1.5480613
McCarty, 2020	1.4147518	1.2988299	1.5410198
Meng, 2020	1.4232006	1.3060794	1.5508245
Mikami, 2020	1.4254555	1.3079046	1.5535716
Mirani, 2020	1.4148802	1.2987224	1.541427
Moon, 2020	1.4160312	1.3000233	1.5423911
Mowla, 2020	1.4159008	1.3000853	1.5420334
Munblit, 2020	1.4157315	1.2994165	1.5424583
Munoz, 2020	1.4121697	1.2966897	1.5379342

Nachega, 2020	1.4223779 1.3055705 1.5496361
Nakeshbandi, 2020	1.4267973 1.3096594 1.5544121
Namendys-Silva, 2020	1.4127643 1.2969795 1.5388856
Narain, 2020	1.4267741 1.3096164 1.5544127
Nicholson, 2020	1.4146136 1.2983321 1.5413096
Nie, 2020	1.406029 1.2912035 1.5310657
Núñez-Gil, 2020	1.4157708 1.2995708 1.5423607
Palaiodimos, 2020	1.4217563 1.3051939 1.5487283
Pan, 2020	1.4125925 1.2967428 1.5387923
Parker, 2020	1.4141377 1.2983069 1.5403026
Perkin, 2020	1.4182899 1.3017601 1.5452511
Petrilli, 2020	1.4239464 1.3066469 1.5517758
Pettit, 2020	1.4267213 1.3101194 1.553701
Poblador-Plou, 2020	1.4229058 1.3053201 1.5510838
Recinella, 2020	1.4212246 1.3046689 1.548193
Reilev, 2020	1.4198674 1.3024342 1.5478889
Rivera-Izquierdo, 2020	1.4176491 1.301313 1.5443856
Rodrıguez-Molinero, 2020	1.4035665 1.2891709 1.5281131
Rosenberg, 2020	1.4131894 1.2970421 1.5397373
Rossi, 2020	1.4164165 1.2998153 1.5434775
Rottoli, 2020	1.4197984 1.3031938 1.5468363
Rozaliyani, 2020	1.4145004 1.2983296 1.5410658
Russo, 2020	1.4240398 1.3070652 1.5514829
Salacup, 2020	1.4208176 1.3044074 1.5476167
Salazar, 2020	1.4101171 1.2948774 1.5356129
Sands, 2020	1.42993 1.3131154 1.5571364
Seiglie, 2020	1.4185754 1.3022965 1.5452365
Shah, 2020	1.4121155 1.2965516 1.5379798
Shang, 2020	1.4250045 1.3082117 1.5522242
Shi, 2020	1.4124031 1.2966639 1.5384731
Shi, 2020	1.4202402 1.3038338 1.5470392
Stony Brook COVID-19	Research Consortium, 2020 1.40791881.2927667 1.5333279
Sun, 2020	1.4207453 1.3044379 1.547423
Talavera, 2020	1.4189814 1.3024817 1.5459015
Tartof, 2020	1.4185152 1.3018947 1.5455824
Tehrani, 2020	1.4184757 1.3020438 1.5453192
Wang, 2020	1.3947774 1.2842842 1.5147767
Wang, 2020	1.4169366 1.3009491 1.5432652
Wang, 2020	1.4196205 1.3029077 1.5467883
Wu, 2020	1.4162004 1.3000072 1.5427787
Wu, 2020	1.4254589 1.3084775 1.5528989

Xie, 2020	1.4142019	1.2984338	1.5402918
Yang, 2020	1.4126209	1.2969178	1.5386463
Yazdanpanah, 2020	1.4077736	1.2926683	1.5331283
Yu, 2020	1.4214995	1.3047308	1.5487185
Zangrillo, 2020	1.412156	1.2967449	1.5378386
Zhou, 2020	1.4105095	1.2950244	1.536293
Combined	1.4180375	1.3021785	1.5442047

Supplementary Figure 4: Influence analysis of hypertension and mortality risk in patients with COVID-19

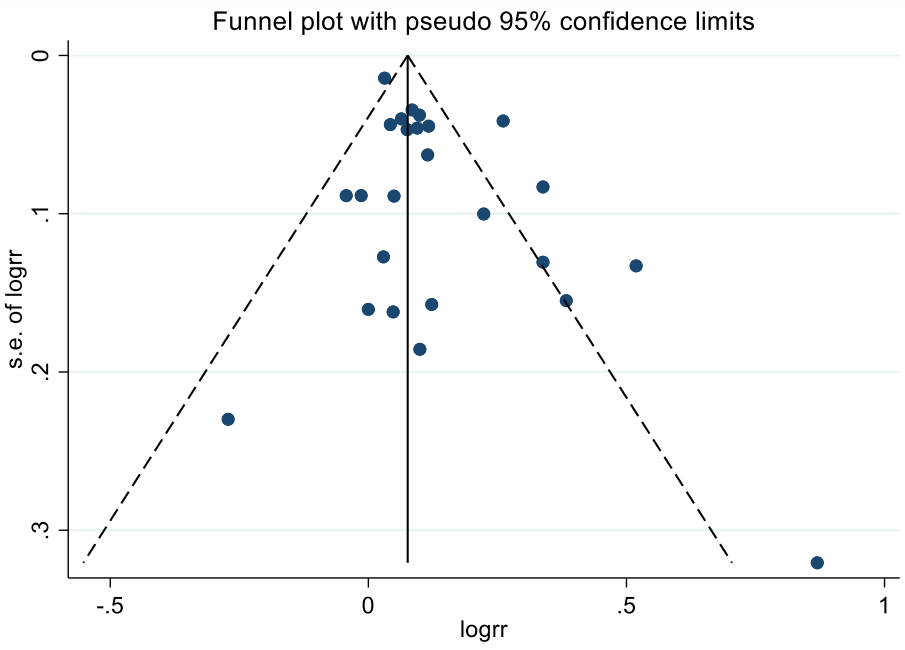


Supplementary Figure 5: Funnel plot of obesity and mortality risk in patients with COVID-19

Study omitted	Estimate	[95% Conf.	Interval]
Almazeedi, 2020	1.4554933	1.3114676	1.6153359
McCarty, 2020	1.4626863	1.3171904	1.6242536
Mccullough, 2020	1.4678108	1.3216922	1.6300832
Salazar, 2020	1.4587231	1.3135937	1.6198866
Arshad, 2020	1.4721547	1.3261496	1.6342347
Stony Brook COVID-19, 2020	1.4625834	1.3163379	1.6250767
Rosenberg, 2020	1.4645751	1.317825	1.6276668
Price-Haywood, 2020	1.4633511	1.3166041	1.6264545
Salacup, 2020	1.4560728	1.3184568	1.6080528
Hashemi, 2020	1.454875	1.3098872	1.6159111
Biscarini, 2020	1.4580365	1.3122839	1.6199776
Fried, 2020	1.4623791	1.3147643	1.6265674
Tehrani, 2020	1.4557939	1.3104337	1.6172782
Kim, 2020	1.4611641	1.3138261	1.6250254
Mendy, 2020	1.4512935	1.3070762	1.6114231
Rodrıguez-Molinero, 2020	1.4539628	1.3089103	1.61509
Crouse, 2020	1.4545934	1.3092545	1.6160662
Hajifathalian, 2020	1.4535341	1.3084871	1.6146597
Berenguer, 2020	1.4570931	1.3097397	1.6210245
Castelnuovo, 2020	1.4555653	1.3092505	1.6182317
Portoles, 2020	1.4520395	1.3071913	1.6129383
Ramos-Rincon, 2020	1.4555131	1.3086935	1.618804
Lombardi, 2020	1.4531474	1.3074876	1.6150345
Lunski, 2020	1.4534943	1.3069357	1.616488
Mallow, 2020	1.4557445	1.3068982	1.6215434
Rossi, 2020	1.4502716	1.3058093	1.6107157
Nakeshbandi, 2020	1.4531265	1.3068702	1.615751
Parker, 2020	1.4498988	1.3056446	1.610091
Docherty, 2020	1.4553313	1.3059231	1.6218332
Mirani, 2020	1.450893	1.3057793	1.6121336
Zhang, 2020	1.4518838	1.3054672	1.6147219
Sands, 2020	1.4500164	1.3040735	1.6122923
Núñez-Gil, 2020	1.4474576	1.3029506	1.6079915
Wollenstein-Betech, 2020	1.4460288	1.2970564	1.6121114
Xie, 2020	1.4454393	1.3012034	1.6056634
Gutierrez, 2020	1.4410919	1.2967353	1.6015188
Sousa, 2020	1.447451	1.3037943	1.6069363
Pettit, 2020	1.4440517	1.2996014	1.6045576
DF de Souza, 2020	1.4449612	1.3009996	1.6048528
Shah, 2020	1.4426155	1.2983586	1.6029004

Imam, 2020	1.4404271	1.295764	1.6012409
Calmes, 2020	1.4430385	1.2988892	1.6031855
Rottoli, 2020	1.4371952	1.2937157	1.5965873
Klang, 2020	1.4298403	1.2871754	1.5883176
Nachega, 2020	1.4382833	1.2949491	1.5974829
Alamdari, 2020	1.4401046	1.2969043	1.5991166
Coppelli, 2020	1.4412719	1.2981442	1.6001803
Giacomelli, 2020	1.4355083	1.2928371	1.5939237
Yazdanpanah, 2020	1.4316087	1.2893935	1.5895095
Gazzaruso, 2020	1.4244035	1.2831527	1.5812036
Mather, 2020	1.4267011	1.2856641	1.5832098
Steinberg, 2020	1.4363017	1.2941382	1.5940822
Escalera-Antezana, 2020	1.4401749	1.2977738	1.5982014
Abohamr, 2020	1.3768033	1.2478951	1.5190276
Combined	1.4483488	1.3049973	1.6074472

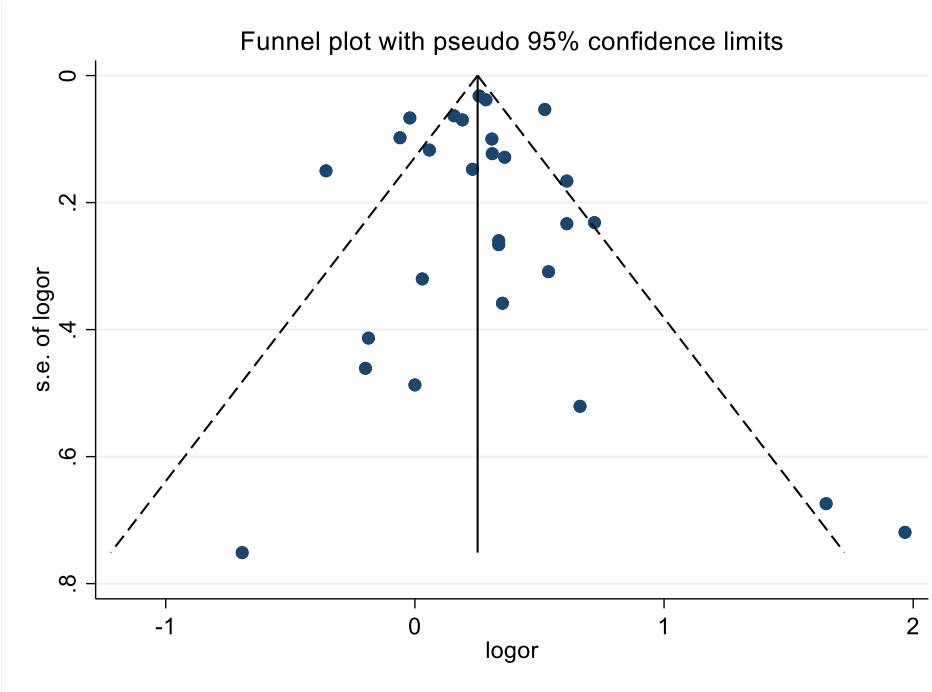
Supplementary Figure 6: Influence analysis of obesity and mortality risk in patients with COVID-19



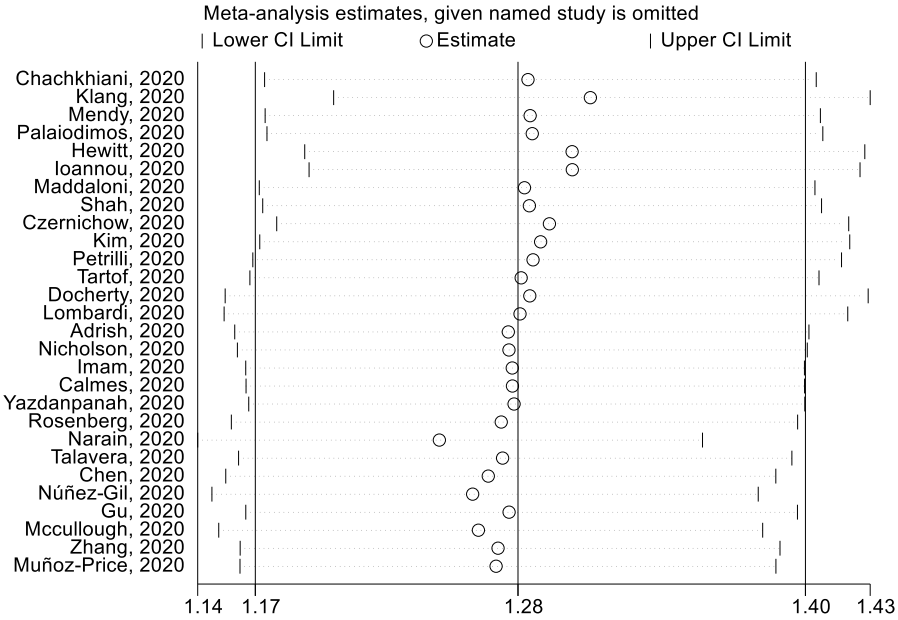
Supplementary Figure 7: Funnel plot of BMI per 5kg/km^2 and mortality risk in patients with COVID-19

Study omitted	Estimate	[95% Conf.	Interval]
Salazar, 2020	1.1262932	1.0773913	1.1774148
Haase, 2020	1.1295877	1.0794542	1.1820494
Li, 2020	1.1284989	1.0781709	1.1811761
Calmes, 2020	1.1250597	1.0753614	1.177055
Hajifathalian, 2020	1.1252736	1.0753136	1.1775547
Ferrando-Vivas, 2020	1.1313956	1.0795149	1.1857696
Anderson, 2020	1.1291318	1.0770124	1.1837733
Seiglie, 2020	1.1241773	1.0744675	1.176187
Lombardi, 2020	1.1259074	1.0754082	1.1787779
Petrilli, 2020	1.1278753	1.0753478	1.1829686
Kabarriti, 2020	1.126538	1.0745648	1.1810251
Santos, 2020	1.1266743	1.0735707	1.1824048
Gupta, 2020	1.1251055	1.0731854	1.1795374
Wang, 2020	1.1251912	1.0726496	1.1803064
Palaiodimos, 2020	1.1231542	1.073586	1.175011
Nakeshbandi, 2020	1.1232269	1.0722753	1.1765996
Klang, 2020 (b)	1.1233695	1.0716629	1.1775709
Gu, 2020	1.1227893	1.0731335	1.1747427
Shah, 2020	1.1186447	1.0692952	1.170272
Czernichow, 2020	1.1065391	1.0623916	1.1525211
Rottoli, 2020	1.116652	1.0682974	1.1671954
Gayam, 2020	1.1111181	1.0643468	1.1599447
Baronio, 2020	1.1171073	1.0688614	1.1675309
Klang, 2020 (a)	1.1115534	1.0657233	1.1593543
Muñoz-Price, 2020	1.1179643	1.0707144	1.1672993
Chachkhiani, 2020	1.1227052	1.0738231	1.1738126
Salazar, 2020	1.1227052	1.0738231	1.1738126
Ghany, 2020	1.1227052	1.0738231	1.1738126
Zimering, 2020	1.1227052	1.0738231	1.1738126
Tartof, 2020	1.1227052	1.0738231	1.1738126
Ioannou, 2020	1.1227052	1.0738231	1.1738126
Recinella	1.1227052	1.0738231	1.1738126
Harmouch, 2020	1.1227052	1.0738231	1.1738126
Combined	1.1227053	1.0738231	1.1738127

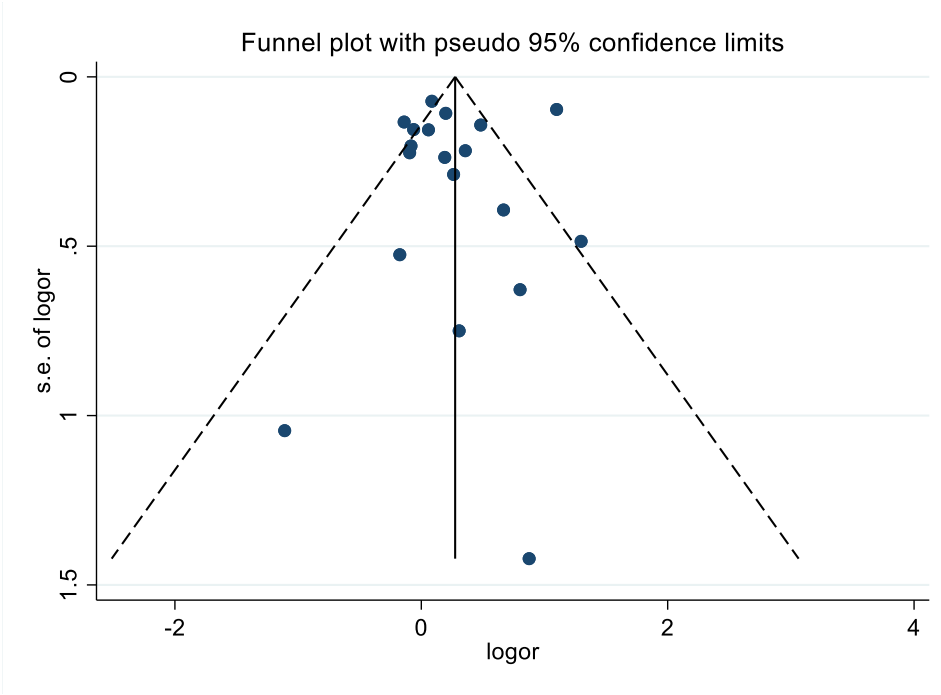
Supplementary Figure 8: Influence analysis of BMI per 5kg/km^2 and mortality risk in patients with COVID-19



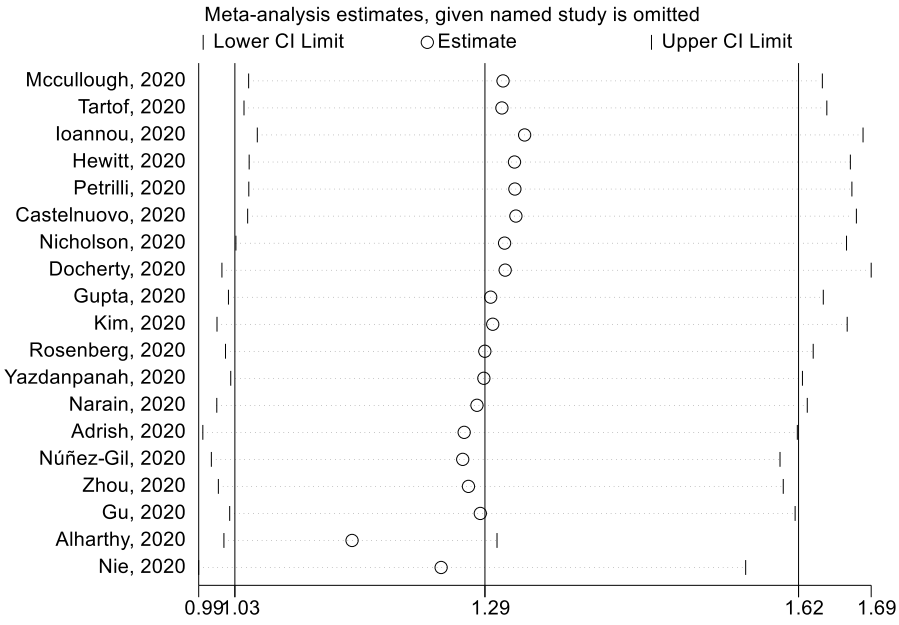
Supplementary Figure 9: Funnel plot of ever smoking vs never smoking and mortality risk in patients with COVID-19



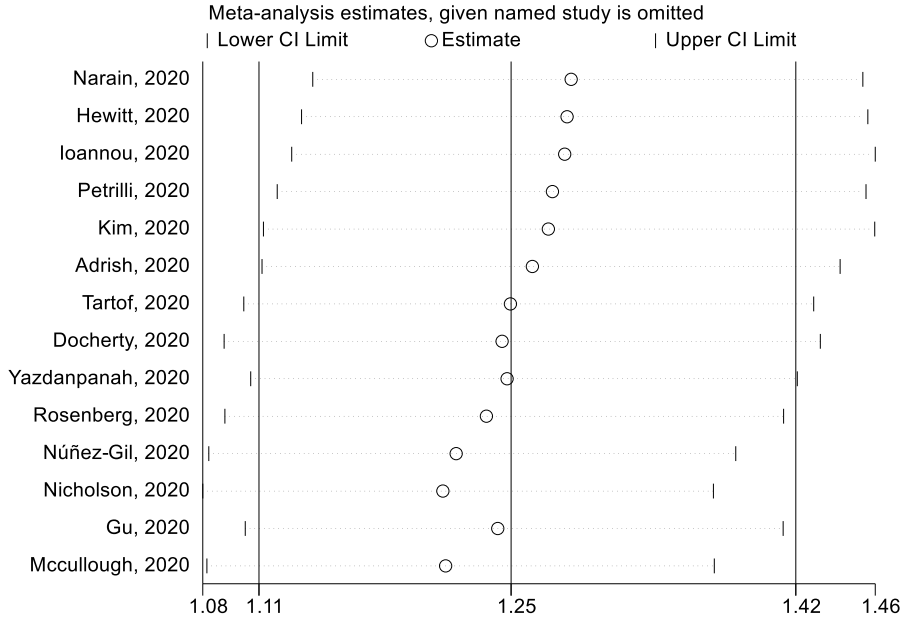
Supplementary Figure 10: Influence analysis of ever smoking vs never smoking and mortality risk in patients with COVID-19



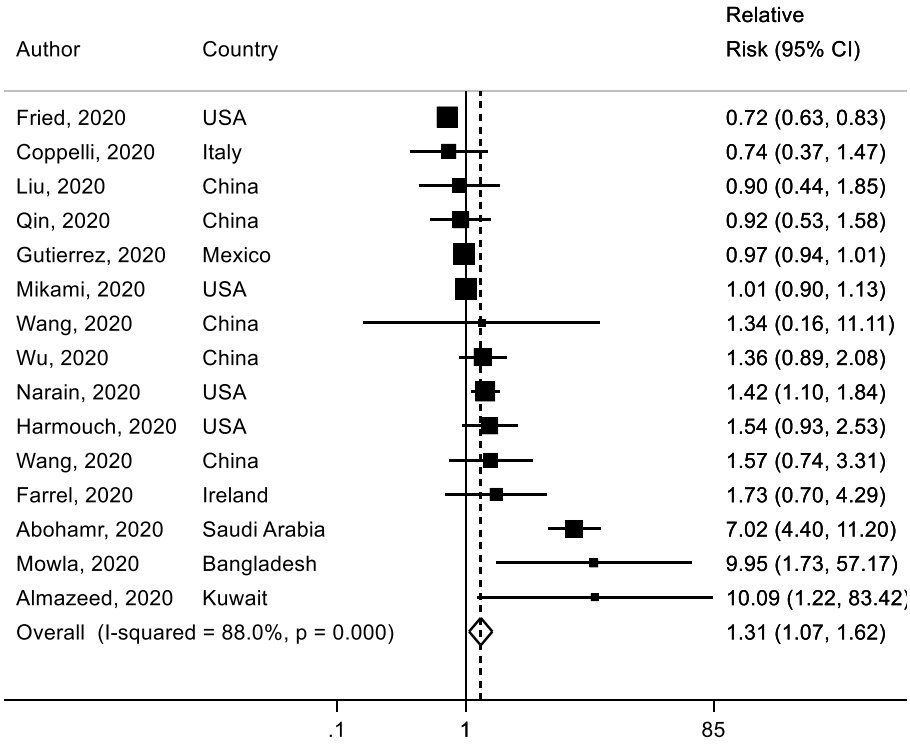
Supplementary Figure 11: Funnel plot of current smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 12: Influence analysis of current smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 13: Influence analysis of former smoking vs never smoking and mortality risk in patients with COVID-19



Supplementary Figure 14: Association between undefined smoking status (versus never smoking) and mortality risk in patients with COVID-19